Next-Step Revision Plan for Integrity-Aware Navigation

From current implementation to a defensible GNSS–LiDAR advisory integrity field

Goal: keep the certified monitoring path conservative, while making the future planning path physically meaningful, queryable, and experimentally defensible.

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Revision Principle: Separate Monitoring from Prediction

Certified / current path

- Input: current FGO snapshot and real measurements.
- Output: current GNSS PL, LiDAR PL, alarm flag.
- Fusion remains conservative:

$$PL_q^{mon} = \max(PL_q^G, PL_q^L)$$

Advisory / future path

- Input: candidate position, map, satellite geometry, current covariance prior.
- Output: queryable predicted PL proxy / integrity margin field.
- Fusion uses information addition:

$$\Lambda_{pred} = \Lambda_{prior} + \Lambda_G(p) + \Lambda_L(p)$$

Key message for the supervisor: monitoring remains conservative and certifiable-ish; planning uses fused advisory information to exploit GNSS–LiDAR complementarity.

Seven Questions → Concrete Next-Step Revisions

Q	Weakness / question	Required revision	Deliverable evidence
1	LiDAR ARAIM PL process, scientific validity, and linear growth.	Decompose LiDAR PL into $\lvert d_f \rvert$, $K_{fa}\sigma_{ss}$, $K_{md}\sigma_f$, b_f. Replace unbounded age risk by bounded/saturated aging; fix target-window size; log each γ component.	PL component plot; static-test drift check.
2	GNSS PL and LiDAR PL have different ranges; how to put them in one dimension?	Keep current monitor conservative in meters: $PL_q^{mon} = \max(PL_q^G, PL_q^L)$. For planning, fuse at information level, not by hand-weighted PL scores.	Separate `PL_mon` and `PL_pred` topics.
3	Predicted GNSS PL is not strict traditional ARAIM PL.	Rename it as *GNSS advisory predicted PL proxy*. Build it from predicted visibility, sky mask/canopy model, $G^\top WG$, and conservative geometry/bias overbounds.	Terminology corrected in code/slides; current-pose consistency plot.
4	Predicted LiDAR PL is too simple; range-only confidence is not scientific.	Replace scalar range confidence with LiDAR observability/FIM field: voxel normals, visibility, map age, degeneracy, TDOP/normal diversity, then compute anisotropic uncertainty.	Corridor / forest anisotropy heatmap.
5	GNSS and LiDAR prediction need one PI output, not multi-source values.	Define unified advisory information: $\Lambda_{pred} = \Lambda_{prior} + \Lambda_G + \Lambda_L$, then PL_{pred}, $IM = AL - PL_{pred}$, and one scalar c_{PI}.	Single PI cost field used by A* and back-end.
6	GNSS and LiDAR are complementary/opposite: open sky vs feature-rich areas.	Do not use fixed source weights. Let the route be selected by fused IM. Add optional source-dropout ablation $(PL_{\setminus G}, PL_{\setminus L})$ to test robustness.	Trade-off route examples; fault-ablation results.
7	A* map, PL field, duplicated computation, and frequency mismatch.	Build a Unified Risk Grid: $\{ESDF, occupancy, AL, PL, IM, age, flags\}$. A* and B-spline query the same cached grid; predictor runs asynchronously with stale/unknown-risk handling.	Runtime breakdown; no duplicated PL query pipeline.

Revised Architecture: Each Issue Maps to One Data-Path Change

Certified current monitor: preserve conservatism

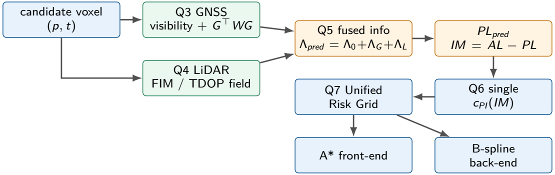
- **Q1:** stabilize LiDAR ARAIM diagnostic terms.
- **Q2:** same meter-domain output for GNSS/LiDAR PL.
- Do not claim future prediction as certified ARAIM.

$$PL_q^{mon} = \max\left(PL_q^{GNSS}, PL_q^{LiDAR}\right)$$

Main risk to fix first

$$\gamma_{age} = \frac{Age}{Age_{ref}} \quad \Rightarrow \quad \gamma_{age} = 1 - e^{-Age/\tau}$$

Unbounded age terms can create artificial LiDAR PL drift.



Planning path: advisory only, but physically meaningful and queryable. The planner receives one scalar integrity cost instead of competing GNSS/LiDAR scores.

Implementation Order Aligned with the Seven Questions

Priority	Modification	Questions solved	Why this order
P0	LiDAR ARAIM diagnostics: PL-term logging, bounded age, fixed target window.	Q1	Must remove artificial PL drift before judging the planner.
P1	Terminology split: *monitored PL* vs *advisory predicted PL proxy*.	Q2, Q3	Prevents overclaiming and makes supervisor explanation defensible.
P2	GNSS predictor upgrade: sky mask / canopy visibility / predicted $G^\top WG$.	Q3	Gives the GNSS field a clear geometry-based meaning.
P3	LiDAR predictor upgrade: voxel-normal FIM, TDOP, degeneracy, map age.	Q4	Replaces range-only confidence with observability.
P4	Advisory FIM fusion and unified PI cost: $\Lambda_G + \Lambda_L \to PL \to IM \to c_{PI}$.	Q5, Q6	Solves source fusion and complementarity without hand-tuned source weights.
P5	Unified Risk Grid for ESDF + AL + PL + IM + age + flags.	Q7	Removes duplicated computation and gives A*/B-spline one consistent interface.
P6	Ablation and runtime experiments: max vs FIM-add, front-only/back-only/full, stale-field behavior.	Q1–Q7	Converts the revision plan into defendable evidence.

Recommended wording: "We do not replace the certified monitor. We keep it conservative, and redesign the planner-side prediction as a fused advisory information field."

Module 1: LiDAR ARAIM PL Stability

Current method

LiDAR ARAIM uses VGICP source–target–level fault hypotheses and snapshot downdate:

$$\Lambda_f = \Lambda_0 - \sum_{F_{j,\ell} \in B(H_f)} \Delta\Lambda_{j,\ell}$$

$$PL^{L}_{q,f} = |e_q^\top d_f| + K_{fa}\sigma_{ss} + K_{md}\sigma_f + b_{q,f}$$

The current block risk score includes RMSE, inlier ratio, condition proxy, and target age.

Problem / risk

The LiDAR PL can show artificial linear growth. The most likely causes are:

- an unbounded age term in $\gamma_{j,\ell}$,
- a growing target-keyframe set under max over hypotheses,
- insufficient logging of the separate PL components.

Why change

If the PL drifts without real degradation, all downstream predictor and planner experiments become contaminated. The monitor may look conservative, but it is no longer physically interpretable.

New scheme

Use a bounded age model and fixed target window:

$$\gamma_{age} = 1 - e^{-Age/\tau} \quad \text{or} \quad \gamma_{age} = \min(Age/A_{ref}, \gamma_{max})$$

$$|T_t| = K \quad \text{fixed sliding target set}$$

Log $|d_f|$, $K_{fa}\sigma_{ss}$, $K_{md}\sigma_f$, b_f and each γ component.

Acceptance check: In static and constant-velocity tests, LiDAR PL should stop showing unbounded linear drift. The RMS drift of PL over a 10 min stationary test should be below a configured threshold.

Module 2: Certified GNSS–LiDAR Fusion

Current method

Stage 1 monitor computes GNSS PL and LiDAR PL independently, then uses conservative fusion:

$$PL_q^{fused} = \max(PL_q^{GNSS}, PL_q^{LiDAR})$$

This keeps fault libraries independent and avoids a joint GNSS–LiDAR fault tree.

Problem / risk

This fusion is overconservative and does not exploit GNSS–LiDAR complementarity. However, replacing it by FIM-add in the monitor can underestimate risk when single-source faults are not fully modeled.

Why change

The current monitor is the safety-critical path. It should remain simple, conservative, real-time, and easy to debug. A joint fault tree scales combinatorially and is not ready for Stage 1.

New scheme

Keep conservative max in the certified/current monitor:

$$PL_q^{mon} = \max(PL_q^{G}, PL_q^{L})$$

Move FIM-level fusion only to the advisory predictor. Explicitly label the two paths in the slides and code comments.

Acceptance check: Current alarm behavior remains unchanged after planner-side upgrades. GNSS-only and LiDAR-only fault injection should still trigger conservative monitor responses.

Module 3: GNSS Future Prediction

Current method

The current future GNSS field uses satellite geometry, visible satellites, and measurement variance models to estimate a position-dependent GNSS risk / PL-like value:

$$\Lambda_G(p,t) = G(p,t)^\top W(p,t) G(p,t)$$

Problem / risk

A future candidate position has no real pseudorange residual and no observed subset solution separation. Therefore it is not a certified ARAIM PL in the strict current-epoch sense.

Why change

Calling it simply "GNSS PL" can be challenged: traditional ARAIM PL depends on actual measurements, fault hypotheses, residuals, and solution separation at the current epoch.

New scheme

Rename and formalize it as an advisory prediction:

GNSS Advisory Predicted PL Proxy

with optional conservative upper-bound terms:

$$\widehat{|d_{q,f}|}(p) \le b_{q,f}(p) + n_\sigma \sigma_{ss,q,f}(p)$$

The output remains compatible with PL/AL, but is explicitly non-certified.

Acceptance check: All documents and topics distinguish current certified PL from future advisory predicted PL proxy. At $p = p_t$, predicted and monitored PL should be numerically close after calibration.

Module 4: LiDAR Future Prediction

Current method

The current LiDAR predictor is mainly a scalar confidence modulator based on local point availability, distance, or crude degeneracy heuristics.

Problem / risk

A scalar confidence cannot represent scan-matching observability. LiDAR localization uncertainty is directional: a corridor may be well constrained laterally but weak along the corridor direction.

Why change

A planner needs a spatially and directionally meaningful LiDAR information field. Otherwise it cannot explain or exploit geometric observability.

New scheme

Upgrade the LiDAR predictor to a lightweight FIM model. First version:

$$\Lambda_L^{trans}(p) \approx \sum_{m \in \mathcal{V}(p)} \frac{1}{\sigma_m^2} n_m n_m^\top \in \mathbb{R}^{3\times 3}$$

where $\mathcal{V}(p)$ is the potentially visible voxel/normal set. Use voxel normal histograms first; add ray-cast visibility later. Long-term target:

$$\Lambda_L^{pose} = \sum_m J_m^\top R_m^{-1} J_m \in \mathbb{R}^{6\times 6}$$

Acceptance check: In a corridor test, predicted uncertainty should be anisotropic: along-corridor PL larger than cross-corridor PL. The field should visually match map geometry.

Module 5: Advisory GNSS–LiDAR FIM Fusion

Current method

The old planner-side design effectively treats GNSS and LiDAR as two PL channels and combines them at the cost level or by max-like logic.

Problem / risk

GNSS and LiDAR are complementary measurements of the same state. A PL-level max loses the benefit of complementary information and can let one source dominate the whole cost field.

Why change

Planning should not search for "GNSS-good" or "LiDAR-good" regions separately. It should search for regions where the total expected localization information is sufficient.

New scheme

Fuse in the information layer:

$$\Lambda_{pred}(p,t) = \Lambda_{prior}(t) + \Lambda_G(p,t) + \Lambda_L(p,t)$$

Then derive the advisory covariance and PL proxy:

$$\Sigma_{pred}(p,t) = \Lambda_{pred}^{-1}, \qquad PL_q^{pred} = K_q \sqrt{e_q^\top \Sigma_{pred}\, e_q} + b_q^{pred}$$

Acceptance check: A/B test `fusion_mode=max` vs `fusion_mode=fim_add`. FIM-add should produce tighter but still bounded predicted PL fields in complementary GNSS/LiDAR scenarios.

Module 6: PI Function Redesign

Current method

The current PI function maps (HPL, VPL) or multi-source PL values into front-end and back-end planning costs. Some previous variants considered balance/floor terms as part of the main objective.

Problem / risk

Multi-source PI inputs make tuning ambiguous. Balance/floor terms can become empirical patches and may obscure the main contribution. Some naive floor formulations even penalize the wrong direction if σ is treated as an error standard deviation.

Why change

The main planner interface should be a single interpretable scalar: predicted integrity margin. Optional robustness terms should be studied through ablation, not baked into the baseline.

New scheme

Default PI cost:

$$IM(p) = AL(p) - PL_{fused}^{pred}(p)$$

$$c_{PI}^{main}(p) = \left[PL_{fused}^{pred}(p) - AL(p) + m\right]_+^2$$

Optional ablation:

$$c_{fault} = \left[PL_{\setminus G} - AL\right]_+^2 + \left[PL_{\setminus L} - AL\right]_+^2$$

Acceptance check: Baseline planner should work with c_{PI}^{main} alone. Balance/fault-robust terms are enabled only for ablation and fault-injection experiments.

Module 7: Unified Risk Grid and Field Aging

Current method

The current system maintains ESDF/occupancy and integrity cost field as separate map-like structures. Field aging may fall back to zero advisory cost when the PL field is stale.

Problem / risk

Separate grids duplicate lookup and synchronization. More importantly, stale integrity information should not mean zero risk; it means unknown risk.

Why change

A* and B-spline optimization should query a consistent multi-layer field. Aging logic must preserve integrity semantics rather than silently hiding stale risk.

New scheme

Build a Unified Risk Grid (URG):

$$URG(v) = \{d_{ESDF}, \rho_{occ}, PL^{pred}_{fused}, AL, IM, age, flags\}$$

Use one interpolator for A* and B-spline. Replace zero fall-back with unknown-risk handling:

$$c_{used} = c_{PI} + c_{unknown}(1 - e^{-\Delta t/\tau})$$

or trigger refresh / slowdown when stale.

Acceptance check: Planner query latency decreases; stale voxels are visible in debugging. The planner should not treat outdated integrity regions as safe free space.

Module 8: Planner Integration

Current method

The front-end A* is biased by integrity cost, and the back-end B-spline optimizer queries the integrity field for gradients or finite differences.

Problem / risk

If the integrity field is noisy, stale, or multi-source inconsistent, A* may generate misleading initial paths and the back-end may fight the front-end.

Why change

The planner should receive one stable, differentiable risk field, not multiple inconsistent signals. This also simplifies parameter tuning and supervisor explanation.

New scheme

Use the unified cost in both stages:

$$e(p_i, p_j) = \|p_i - p_j\| \left(1 + \lambda_{obs} c_{obs}(p_j) + \lambda_{PI} c_{PI}(p_j)\right)$$

Back-end:

$$J = J_{smooth} + J_{obs} + J_{dyn} + \lambda_{PI} \sum_k c_{PI}(c_k)$$

Use finite-difference gradients first; later replace with analytic field gradients.

Acceptance check: Front-only, back-only, and full planner variants should be compared. Full should reduce max predicted PL / improve minimum IM without increasing collision risk.

Module 9: Estimator Information Interface

Current method

The estimator currently provides pose/covariance snapshots and LiDAR block summaries. In some paths, pose covariance is inverted to approximate the current information matrix.

Why change

The predictor and ARAIM should share the same statistical basis. Source-wise FIM extraction improves interpretability, calibration, and debugging.

Problem / risk

iSAM/GTSAM marginal covariance can be optimistic under linearization, correlated LiDAR residuals, wrong noise models, robust losses, and marginalization priors. A pure covariance inversion may not expose source-wise information.

New scheme

Add an optional information export interface:

$$\Lambda_0 = \Lambda_{prior} + \Lambda_{IMU} + \Lambda_G + \Lambda_L$$

$$\Lambda_G = \sum_s J_s^\top R_s^{-1} J_s, \qquad \Lambda_L = \sum_{j,\ell} J_{j,\ell}^\top R_{j,\ell}^{-1} J_{j,\ell}$$

Keep covariance inflation / floor for robust overbounding:

$$\Sigma_{used} = \eta \Sigma_{iSAM} + \sigma_{floor}^2 I$$

Acceptance check: Logs should show source-wise information contribution, covariance inflation factor, and consistency against truth-pose baselines.

Module 10: Validation and Evidence

Current method

Current demonstrations can show that the UAV flies, but visual inspection alone cannot prove integrity-aware benefits.

Problem / risk

Integrity-aware planning avoids tail events. In normal flights, baseline and proposed planner may look similar. Without stress tests, the contribution is hard to defend.

Why change

The next version must connect each algorithmic change to a measurable experiment. The evaluation should prove correctness, prediction consistency, planning benefit, and real-time feasibility.

New scheme

Core experiment set:

- ARAIM: truth-pose consistency, Stanford-like diagram, GNSS/LiDAR fault injection.
- Predictor: self-consistency at current pose, corridor anisotropy, FIM-add vs max.
- Planner: front-only/back-only/full ablation, PI weight sweep.
- System: Monte Carlo tail statistics, latency breakdown, real-world replay.

Acceptance check: At least one figure per claim: PL-vs-error diagram, spatial PL slice, trajectory over PL heatmap, time-series PL/AL/error, tail CDF.

Proposed Implementation Roadmap

Phase	Main task	Exit condition
0	Fix LiDAR PL drift	Static / constant-velocity PL no longer grows linearly.
1	Rename and separate certified vs advisory quantities	All topics/slides distinguish monitored PL and predicted PL proxy.
2a	Build LiDAR 3×3 FIM predictor	Corridor anisotropy is visible in predicted PL field.
2b	Add advisory FIM fusion	`fim_add` and `max` can be A/B tested.
3	Simplify PI function	Planner uses one PL_{fused}^{pred}/AL or IM scalar.
4	Build URG map interface	A* and B-spline use the same grid query API.
5	Optimize runtime	Predictor refresh below target budget; planner remains 10 Hz.
6	Run experiments and ablations	Module-level and end-to-end evidence ready for supervisor review.

One-Slide Summary for Discussion

What is weak now

- LiDAR PL can drift due to age/max-set artifacts.
- Future GNSS "PL" is not strict ARAIM PL.
- LiDAR predictor is too scalar and not observability-aware.
- PL-level max cannot exploit GNSS–LiDAR complementarity for planning.
- PI cost receives inconsistent multi-source signals.
- Stale field fallback can hide unknown risk.

What we will change

- Stabilize LiDAR PL with bounded age and fixed target window.
- Keep monitor conservative; use FIM-add only in advisory predictor.
- Rename future outputs as predicted PL proxies.
- Upgrade LiDAR predictor to FIM / observability field.
- Feed planner one unified IM or PL/AL scalar.
- Merge maps into URG with unknown-risk aging.

Final architecture:

Certified monitor (max PL) $\|$ Advisory predictor $(\Lambda_G + \Lambda_L) \rightarrow IM = AL - PL \rightarrow Planner$